

App Dev Project Report

1. Student Details

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About Me: I am Rahul Pathak at IIT Madras BS Degree program with a deep interest in data science. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Trekking Management Application

Project Statement:

Develop a Trekking Management System using Flask, Vue.js, SQLite, Redis, and Celery. The application supports Administrator, Staff, and Trekker roles with secure authentication, trek management, booking, reporting, caching, and background jobs.

Approach:

The project follows a REST API architecture. Flask provides backend APIs, SQLAlchemy manages the SQLite database, Vue.js is used to build a Single Page Application, Redis provides caching, and Celery executes background jobs such as daily reminder emails, monthly reports, and CSV export.

3. AI / LLM Declaration

AI assistance (ChatGPT) was used for understanding concepts, debugging, code review, validation improvements, documentation, and project structure review. Approximately 20-25% of the development involved AI-assisted suggestions. Final implementation, integration, testing, and debugging were completed manually after understanding the generated code.

4. Frameworks and Libraries Used

Technology	Purpose
Flask	Backend REST API
SQLAlchemy	ORM
SQLite	Database
Vue.js 3	Single Page Application
Vue Router	Routing

Bootstrap 5	UI
Redis	Caching
Celery	Background Jobs
PyJWT	Authentication
Werkzeug Security	Password Hashing
Flask-CORS	CORS
python-dotenv	Environment Variables

5. Database Schema / ER Diagram

Tables

1. User - Stores details of all users in the system, including Administrators, Staff members, and Trekkers. It contains user information such as ID, full name, email, password, contact number, role, and account status.

2. Trek - Stores trekking event details including trek name, location, difficulty level, duration, total and available slots, trek status, start date, end date, description, and the staff member assigned to manage the trek.

3. Booking - Stores trek booking information made by trekkers. It maintains the relationship between users and treks, including booking date, booking status, and prevents duplicate bookings using a unique constraint.

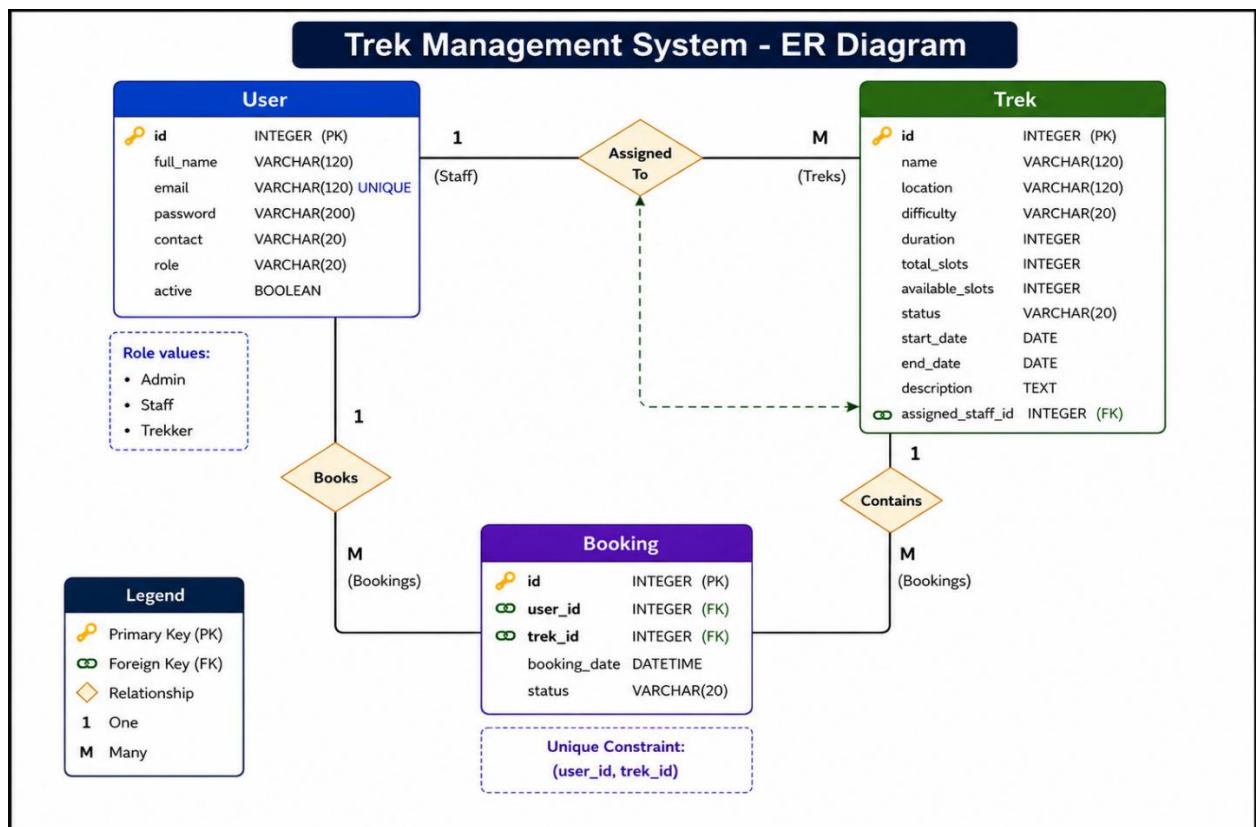
Relationships

- **One-to-Many → User (Staff) → Trek**
 - One staff member can be assigned to manage multiple treks.
- **One-to-Many → User (Trekker) → Booking**
 - One trekker can make multiple trek bookings.
- **One-to-Many → Trek → Booking**
 - One trek can have multiple bookings from different trekkers.

Database Constraints

- **Primary Keys:** User.id, Trek.id, Booking.id

- **Foreign Keys:**
 - Trek.assigned_staff_id → User.id
 - Booking.user_id → User.id
 - Booking.trek_id → Trek.id
- **Unique Constraint:** (user_id, trek_id) in the Booking table to prevent duplicate bookings for the same trek.



6. API Resource Endpoints

Authentication:

POST /api/register

POST /api/login

Admin:

GET/POST /api/admin/treks

PUT/DELETE /api/admin/treks/<id>

PUT /api/admin/treks/<id>/assign

GET/POST /api/admin/staff

PUT /api/admin/staff/<id>/toggle

GET /api/admin/users

PUT /api/admin/users/<id>/toggle

GET /api/admin/bookings

GET /api/admin/dashboard

GET /api/admin/stats

Staff:

GET /api/staff/dashboard

GET /api/staff/treks

PUT /api/staff/treks/<id>

GET /api/staff/treks/<id>/participants

PUT /api/staff/treks/<id>/mark

Trekker:

GET /api/treks

POST /api/bookings

GET /api/bookings

PUT /api/bookings/<id>/cancel

GET/PUT /api/profile

POST /api/export-history

7. Project Features

- Role-based Authentication
- Trek Management
- Staff Management
- Booking Management
- Search and Filter
- Redis Caching
- Daily Reminder Emails
- Monthly Reports

- CSV Export
- Responsive SPA using Vue.js

8. Video Presentation

Google Drive Link:

<https://drive.google.com/file/d/15Niazv0VkPZb1GILnrJv5PJggMN7nfhZ/view?usp=sharing>